4.8 静态随机存储器SRAM——存储器结构及其芯片实例

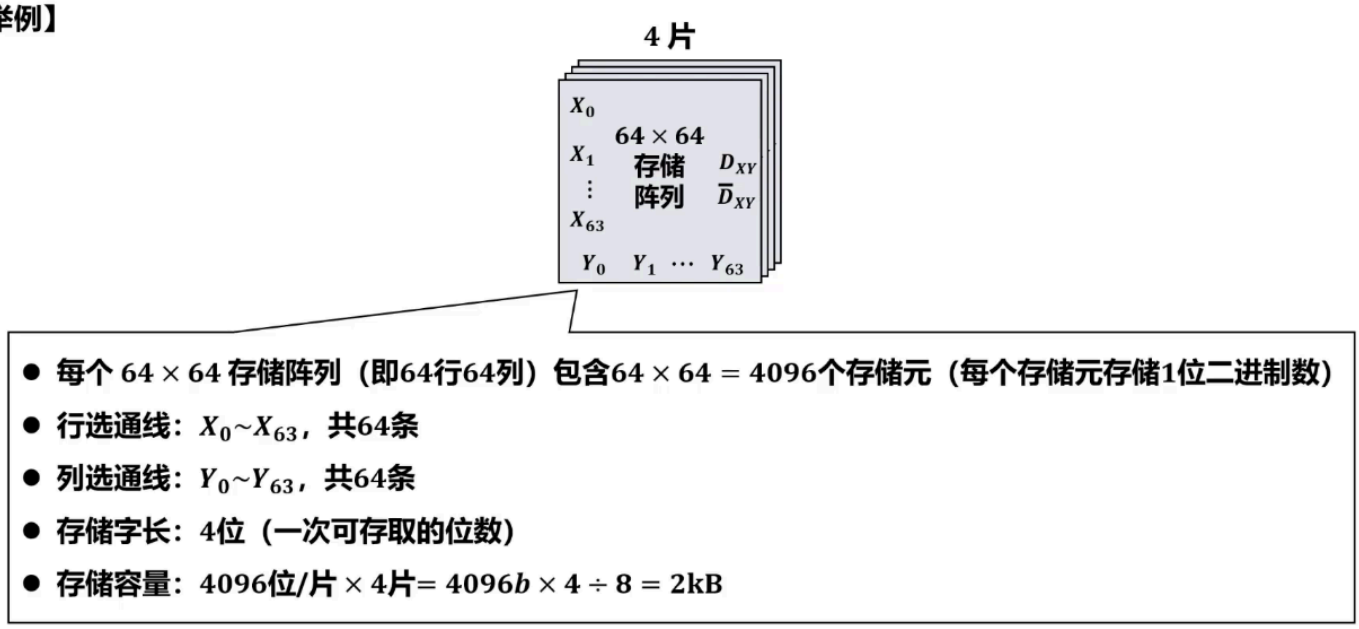
考纲定位：计算机组成原理 · 第 4 章 存储器（4.8）｜重要度 ★★★★★｜常见考法：选择题/大题（芯片引脚、CS/WE、读写控制、地址线与数据线、芯片扩展）

一句话核心：SRAM 芯片 = 存储阵列 + 行/列译码 + I/O 控制；片选 CS 选芯片，写允许 WE 选读/写

一、存储器结构

1.1 规格介绍

【举例】

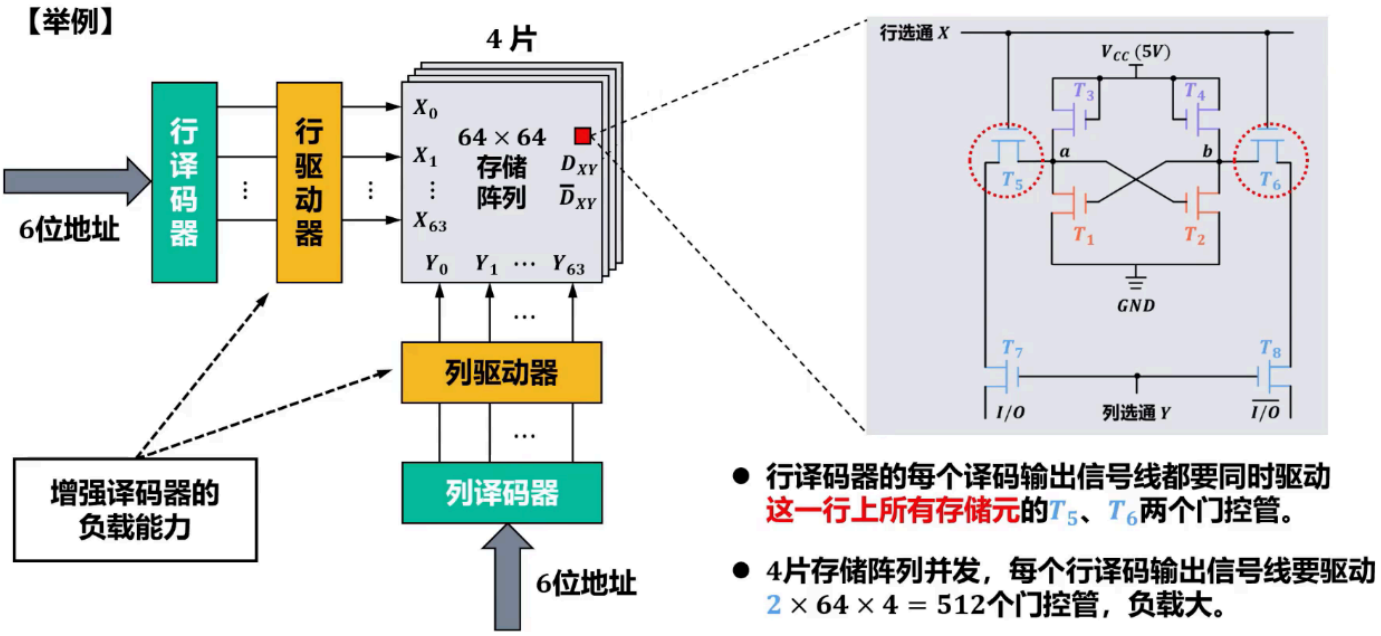


提炼

- 芯片规格常用“字数 × 字长”表示，如 $1K \times 4$ 表示 1024 个存储字，每个字 4 位
- 地址线位数由字数决定：$1K = 2^{10}$，需要 10 根地址线
- 数据线位数等于字长：4 位字长需要 4 根数据线
- 总容量 $= 1K \times 4 = 4K$ 位 $= 512B$

1.2 具体结构

行译码器

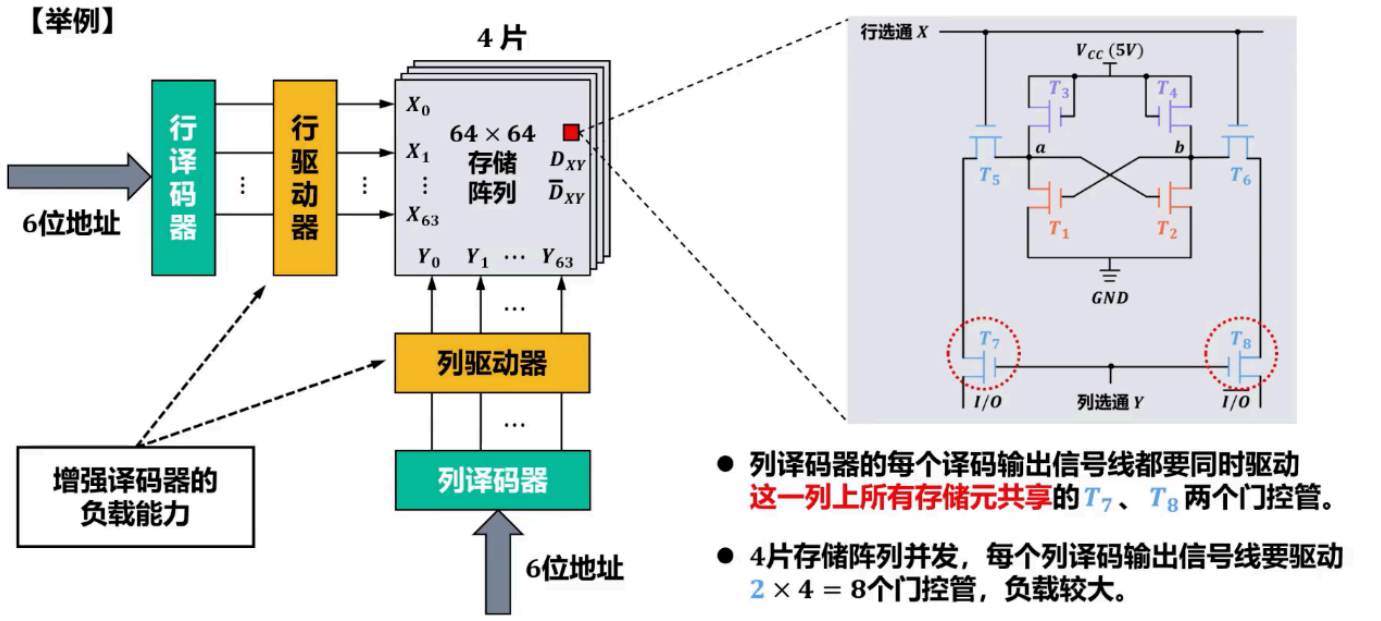


- **行译码器的每个译码输出信号线都要同时驱动这一行上所有存储元的T_5、T_6两个门控管。**
- **4片存储阵列并发，每个行译码输出信号线要驱动$2 \times 64 \times 4 = 512$个门控管，负载大。**

- 行译码器把行地址译成行选择信号，选中存储阵列中的一行
- 行地址位数由“行数”决定

列译码器

静态随机存取存储器SRAM —— 存储器结构

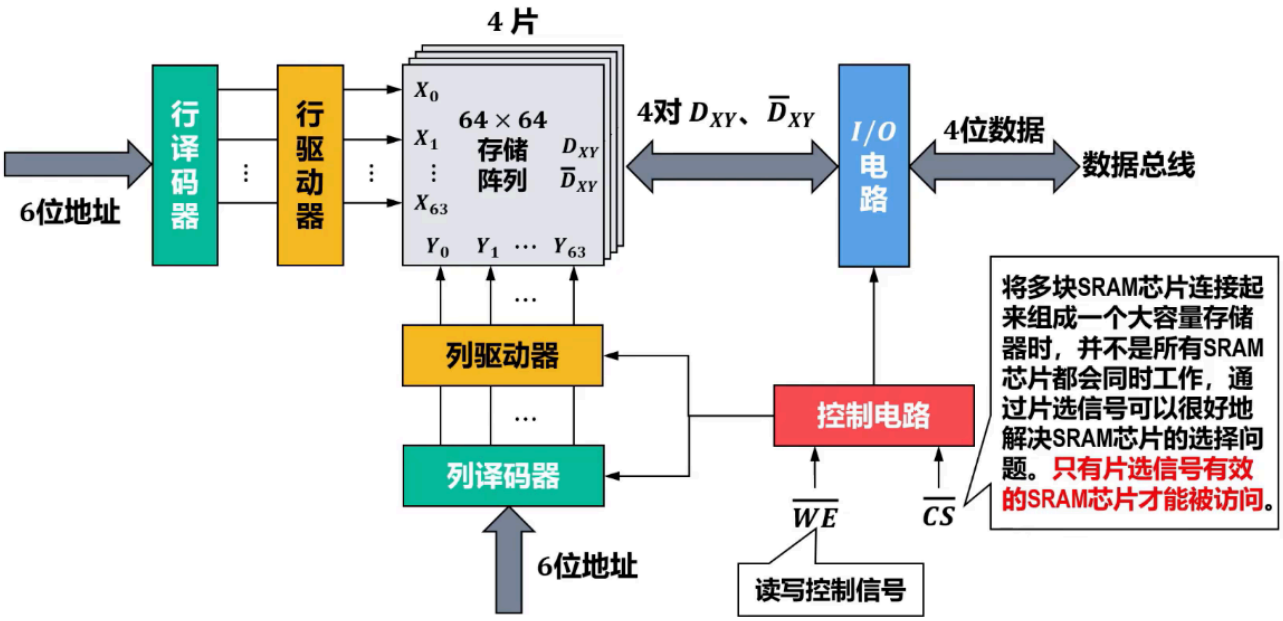


- **列译码器的每个译码输出信号线都要同时驱动这一列上所有存储元共享的T_7、T_8两个门控管。**
- **4片存储阵列并发，每个列译码输出信号线要驱动$2 \times 4 = 8$个门控管，负载较大。**

- 列译码器把列地址译成列选择信号，从选中行中选出目标列
- 行、列交叉选中一个存储单元

I/O 及控制设备

【举例】

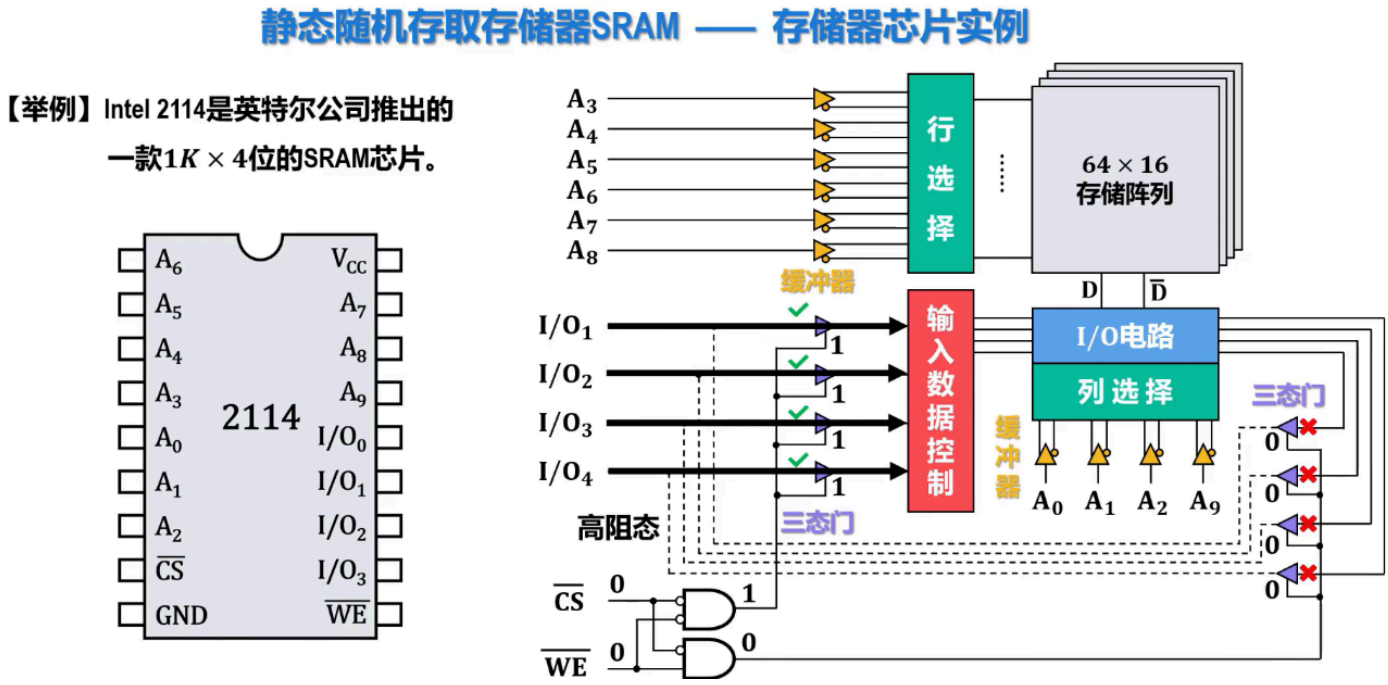


- 数据输入/输出缓冲器连接数据总线
- 读写控制逻辑接收 CS、WE 等信号，决定数据流向
- 三态门在芯片未被选中时输出高阻态，避免总线冲突

二、芯片实例

以 Intel 2114 为例：$1K \times 4$ 的 SRAM 芯片

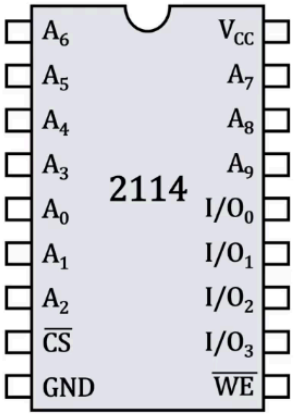
写入操作：CS 有效、WE 有效

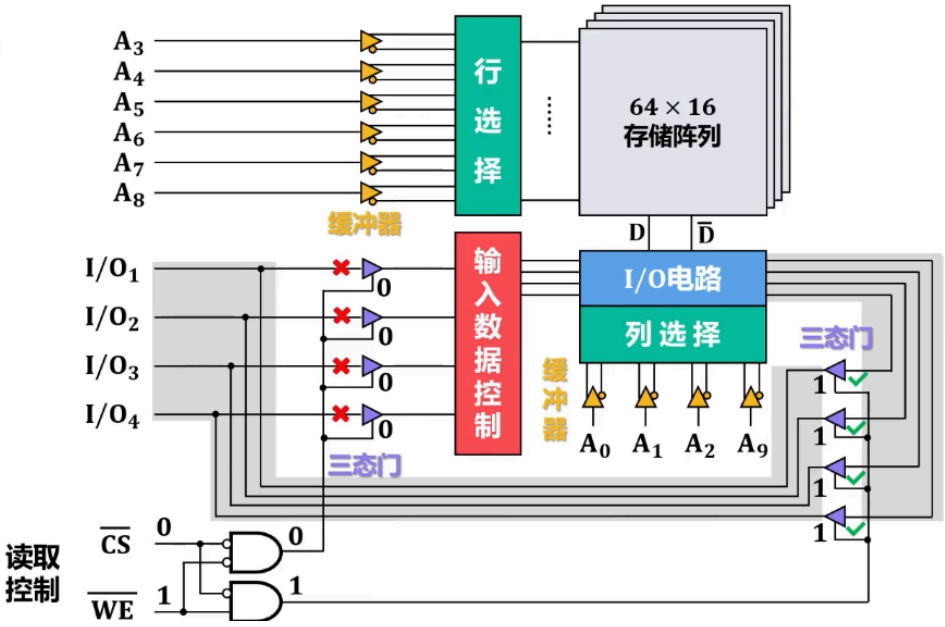


读操作：CS 有效、WE 无效

静态随机存取存储器SRAM —— 存储器芯片实例

【举例】Intel 2114是英特尔公司推出的一款$1K \times 4$位的SRAM芯片。



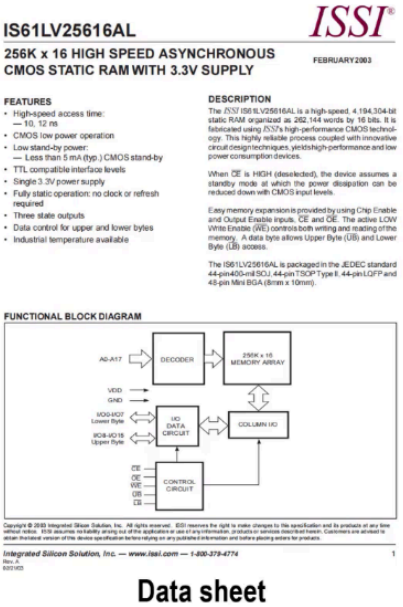


正确控制表

CS	WE	操作	数据线
0（有效）	0（有效）	写	CPU → 芯片
0（有效）	1（无效）	读	芯片 → CPU
1（无效）	×	未选中	高阻态

注：本图中 CS、WE 均为低电平有效；不同芯片的极性可能不同，做题先看引脚符号上的非号/真值表

功能框架


ISSI

IS61LV25616AL

256K x 16 HIGH SPEED ASYNCHRONOUS CMOS STATIC RAM WITH 3.3V SUPPLY

FEBRUARY2003

FEATURES

- High-speed access time: — 10, 12 ns
- CMOS low power operation
- Low stand-by power: — Less than 5 mA (typ.) CMOS stand-by
- TTL compatible interface levels
- Single 3.3V power supply
- Fully static operation: no clock or refresh required
- Three state outputs
- Data control for upper and lower bytes
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DESCRIPTION

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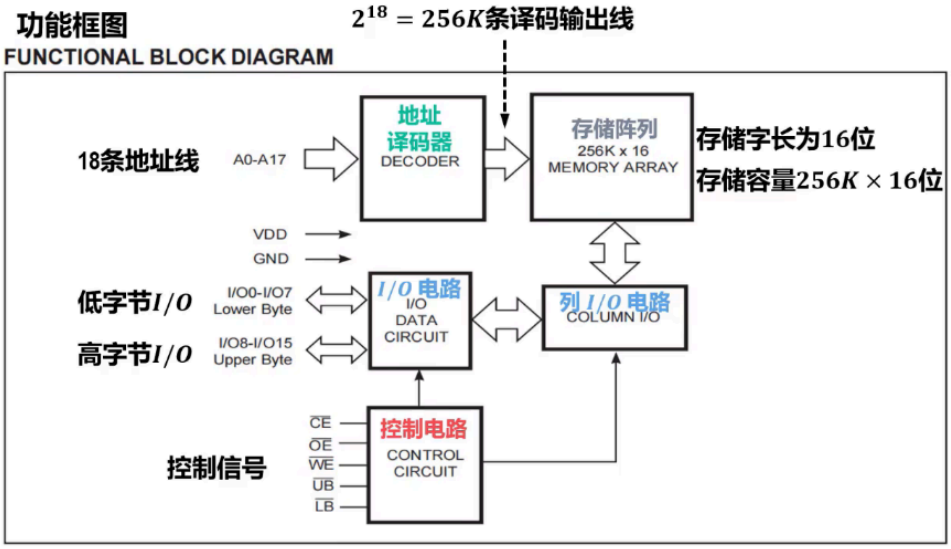
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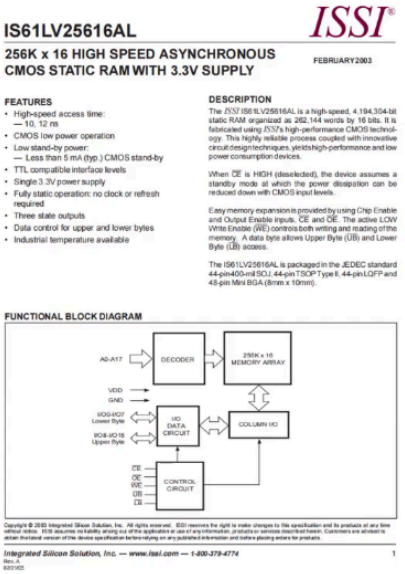
FUNCTIONAL BLOCK DIAGRAM

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Data sheet



真值表



IS61LV25616AL

ISSI®

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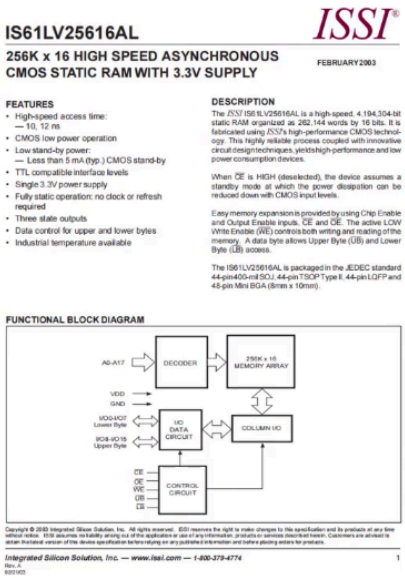
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FUNCTIONAL BLOCK DIAGRAM

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Data sheet

读写周期



IS61LV25616AL

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真值表 TRUTH TABLE

写使能	芯片使能	输出使能	低位字节控制	高位字节控制

High-Z：高阻态

D_{OUT}：数据输出

D_{IN}：数据输出

模式 Mode	控制引脚 $\overline{WE}$	$\overline{CE}$	$\overline{OE}$	$\overline{LB}$	$\overline{UB}$	I/O PIN I/O0-I/O7	I/O引脚 I/O8-I/O15	V_{DD} Current
Not Selected 未选中	X	H	X	X	X	High-Z	High-Z	I_{SB1}, I_{SB2}
Output Disabled 输出禁止	H	L	H	X	X	High-Z	High-Z	I_{CC}
	X	L	X	H	H	High-Z	High-Z	
Read 读操作	H	L	L	L	H	D_{OUT}	High-Z	I_{CC}
	H	L	L	H	L	High-Z	D_{OUT}	
	H	L	L	L	L	D_{OUT}	D_{OUT}	
Write 写操作	L	L	X	L	H	D_{IN}	High-Z	I_{CC}
	L	L	X	H	L	High-Z	D_{IN}	
	L	L	X	L	L	D_{IN}	D_{IN}	

X：高电平或低电平都无影响

H：高电平

L：低电平

每个型号的SRAM芯片有其特定的读写周期特性，只有按照其读写周期去访问SRAM芯片，才能保证读写操作的正确性。

静态随机存取存储器SRAM —— 存储器芯片实例

IS61LV25616AL ISSI®

256K x 16 HIGH SPEED ASYNCHRONOUS CMOS STATIC RAM WITH 3.3V SUPPLY

FEBRUARY 2003

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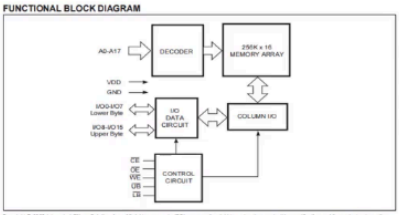
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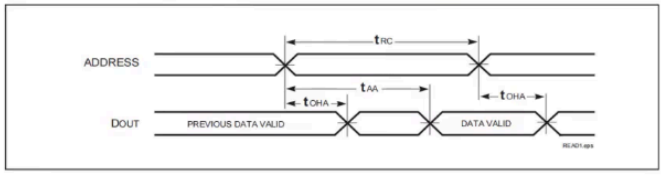


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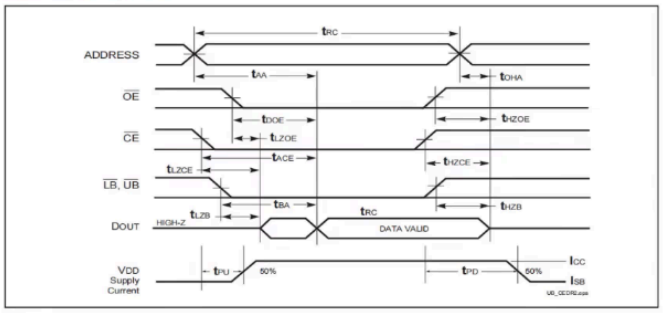
读周期 （模式 1 ）

READ CYCLE NO. 1[1,2] (Address Controlled) ($\overline{CE}$ = $\overline{OE}$ = V_{IL}, $\overline{UB}$ or $\overline{LB}$ = V_{IL})



读周期 （模式 2 ）

READ CYCLE NO. 2[1,3]



Notes:
1. $\overline{WE}$ is HIGH for a Read Cycle.
2. The device is continuously selected. $\overline{OE}$, $\overline{CE}$, $\overline{UB}$, or $\overline{LB}$ = V_{IL}.
3. Address is valid prior to or coincident with $\overline{CE}$ LOW transition.

电源特性

静态随机存取存储器SRAM —— 存储器芯片实例

IS61LV25616AL ISSI®

256K x 16 HIGH SPEED ASYNCHRONOUS CMOS STATIC RAM WITH 3.3V SUPPLY

FEBRUARY 2003

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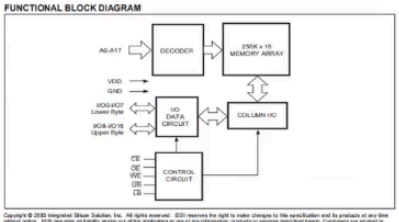
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Data sheet

工作范围

OPERATING RANGE

Range	Ambient Temperature	V_{DD} 10ns	V_{DD} 12ns
Commercial	0°C to +70°C	3.3V +10%, -5%	3.3V ± 10%
Industrial	–40°C to +85°C	3.3V +10%, -5%	3.3V ± 10%

直流电气特性

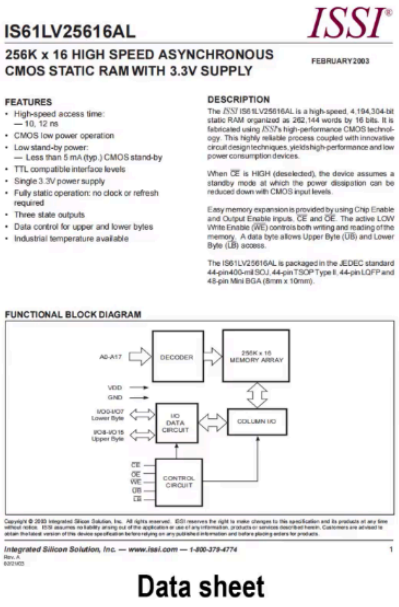
DC ELECTRICAL CHARACTERISTICS (Over Operating Range)

Symbol	Parameter	Test Conditions		Min.	Max.	Unit
V_{OH}	Output HIGH Voltage	V_{DD} = Min., I_{OH} = –4.0 mA		2.4	—	V
V_{OL}	Output LOW Voltage	V_{DD} = Min., I_{OL} = 8.0 mA		—	0.4	V
V_{IH}	Input HIGH Voltage			2.0	V_{DD} + 0.3	V
V_{IL}	Input LOW Voltage[1]			–0.3	0.8	V
I_{LI}	Input Leakage	GND ≤ V_{IN} ≤ V_{DD}	Com. Ind.	–2 –5	2 5	µA
I_{LO}	Output Leakage	GND ≤ V_{OUT} ≤ V_{DD} Outputs Disabled	Com. Ind.	–2 –5	2 5	µA

Notes:
1. V_{IL} (min.) = –2.0V for pulse width less than 10 ns.

芯片封装

静态随机存取存储器SRAM —— 存储器芯片实例



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ISSI

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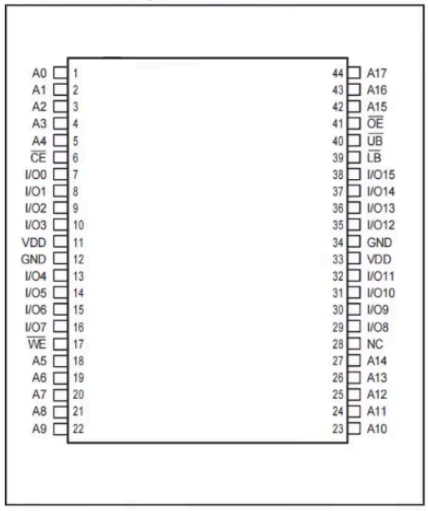
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引脚配置

PIN CONFIGURATIONS

44-Pin TSOP (Type II) and SOJ



PIN DESCRIPTIONS

A0-A17	Address Inputs
I/O0-I/O15	Data Inputs/Outputs
$\overline{CE}$	Chip Enable Input
$\overline{OE}$	Output Enable Input
$\overline{WE}$	Write Enable Input
$\overline{LB}$	Lower-byte Control (I/O0-I/O7)
$\overline{UB}$	Upper-byte Control (I/O8-I/O15)
NC	No Connection
VDD	Power
GND	Ground

提炼

- 真值表用于确认 CS/WE 的有效电平与操作对应关系
- 读写周期给出地址、CS、WE、数据的时序关系；408 主要考“谁先有效、谁后撤销”
- 电源、封装属于了解内容，不是计算重点

三、易错总结

- **CS 与 WE 不要混**：CS 选芯片，WE 选读/写
- **低电平有效要看符号**：$\overline{CS}$、$\overline{WE}$ 上方的横线表示低有效
- **地址线位数由字数决定**：$1K$ 字需要 10 位，不是 4 位
- **数据线位数由字长决定**：4 位字长需要 4 根数据线
- **未选中时数据线高阻态**：多片芯片共享数据总线时，未选中芯片不能驱动总线
- **读操作是非破坏性的**：SRAM 读出后内容不变

知识关联：SRAM 存储元（4.6 静态随机存储器SRAM——存储元）；SRAM 扩展（4.7 静态随机存储器SRAM——存储元扩展和存储阵列扩展）；主存扩展（4.13 主存的扩展及其与CPU的连接——位扩展）；地址译码（4.5B 用于地址译码的译码结构）